**FOUNDATIONS OF ARTIFICIAL INTELLIGENCE**

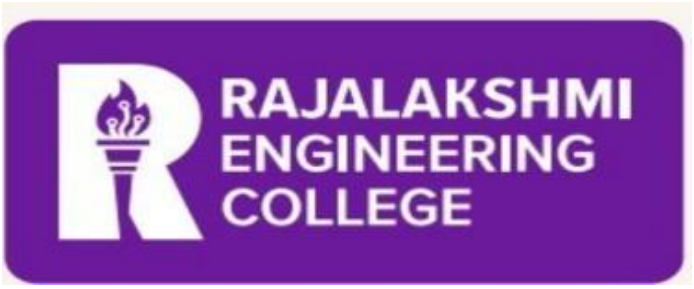
# AI-BASED SMART BLOOD DONOR–RECIPIENT MATCHING & EMERGENCY DISPATCH SYSTEM

## A PROJECT REPORT

Submitted by

**240701616 – RITTVIK S**

**Academic Year 2026–2027**



# BONAFIDE CERTIFICATE

Certified that this project report “AI-Based Smart Blood Donor–Recipient Matching & Emergency Dispatch System” is the bonafide work of “Rittvik S” who carried out the project work under my supervision.

**SIGNATURE OF THE FACULTY INCHARGE**

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# ABSTRACT

Delays in finding a suitable blood donor remain a serious problem in emergency healthcare. In many hospitals, donors are still traced through phone calls, social-media appeals and personal contacts, which wastes valuable time when a patient needs blood urgently. This project proposes an AI-based smart blood donor–recipient matching and emergency dispatch system that brings this process into one platform. The system keeps a registry of donors and, when an emergency request is raised, filters donors using ABO and Rh compatibility rules and donation-eligibility constraints. The remaining candidates are ranked by a weighted score that combines compatibility grade, travel distance and donor availability, where availability is designed to be predicted by a Random Forest classifier. Travel distance is computed with the A* heuristic search algorithm over a road network, and the top-ranked donors are alerted and dispatched. A prototype tested on a synthetic city grid showed that A* returned the same shortest routes as uniform-cost search while expanding about 70% fewer nodes. The system is a decision-support tool and does not replace clinical screening.

# 1 INTRODUCTION

Blood transfusion is a life-saving treatment used in accidents, surgeries, childbirth complications, cancer care and blood disorders. Blood cannot be manufactured artificially and has a limited shelf life, so hospitals depend on voluntary donors and on blood banks for a continuous supply. When the required blood group is not available in stock, the hospital or the patient's family must find a compatible donor quickly.

In practice this search is largely manual. Relatives call known donors one after another, messages are forwarded on social media, and nobody can easily tell whether a contacted donor has the correct blood group, is medically eligible to donate, or is close enough to reach the hospital in time. Every wasted call adds to the delay.

This project addresses the problem by proposing an AI-based smart blood donor–recipient matching and emergency dispatch system. Instead of contacting donors blindly, the system applies compatibility rules, eligibility rules, route search and a learned availability estimate to produce a ranked list of donors, and then coordinates their alerts and dispatch from a single interface.

## 1.1 Need for the System

- Emergency requests are time-critical, and manual donor tracing is slow and unreliable.
- Donor information is scattered across contact lists, messaging groups and paper registers.
- A donor who is compatible may still be ineligible (donated recently) or too far away.
- Rare blood groups, such as the negative groups, make the search considerably harder.
- Automated ranking and clear status tracking help hospitals act with confidence.

## 1.2 Aim

The aim is to develop an intelligent decision-support system that matches blood recipients with suitable donors and coordinates emergency dispatch, using compatibility rules, machine learning for donor availability, and A* search for routing.

## 1.3 Objectives

1. Maintain a registry of donors with blood group, location, last donation date and contact details.
2. Accept emergency requests that specify blood group, number of units, hospital and urgency.
3. Filter donors using ABO and Rh compatibility rules.
4. Exclude donors who are inside the minimum interval between donations.
5. Predict donor availability using a Random Forest classification model.
6. Compute travel distance from each donor to the hospital using A* search.
7. Rank donors with a weighted score and alert the best candidates first.
8. Track donor responses and update records after each donation.

## 1.4 Scope

The project covers donor registration, request handling, compatibility and eligibility filtering, AI-based ranking, route computation and dispatch coordination. It supports human decision-making by hospital staff. It does not perform medical screening, cross-matching or blood testing, which remain the responsibility of qualified clinical personnel and licensed blood banks.

## 1.5 Organization of the Report

Chapter 2 reviews related work. Chapter 3 presents the proposed system and its architecture. Chapter 4 describes the individual modules. Chapter 5 explains the implementation and results, and Chapter 6 gives the conclusion and future work. References are listed in Chapter 7.

# 2 LITERATURE REVIEW

The proposed system combines ideas from blood-group immunology, search algorithms, machine learning and assignment optimisation. The main approaches considered in this project are summarised below.

| Area | Approach considered in this project |
|---|---|
| Blood donation management | Registries and blood-bank portals that store donor and stock information; donor mobilisation depends on voluntary donation (World Health Organization, 2010). |
| Blood group compatibility | Rule-based matching on ABO and Rh antigens to decide which donor groups a recipient can receive. |
| Shortest-path search | Dijkstra's algorithm (1959) and the heuristic A* algorithm (Hart, Nilsson and Raphael, 1968). |
| Machine learning | Ensemble classification with Random Forests (Breiman, 2001) as implemented in scikit-learn (Pedregosa et al., 2011). |
| Assignment optimisation | The Hungarian method (Kuhn, 1955) for assigning donors to several simultaneous requests. |
| Language understanding | Pre-trained transformer models such as BERT (Devlin et al., 2019) for reading free-text emergency messages. |

## 2.1 Blood Donation Management

The World Health Organization promotes a stable supply of safe blood based on regular, voluntary, unpaid donors (World Health Organization, 2010). Managing such donors requires reliable records of who can donate, when they last donated and how to reach them. Most registry-based systems, however, mainly store and display this information and leave the task of choosing and contacting donors to the user.

## 2.2 Blood Group Compatibility

A recipient's red cells carry or lack the A, B and Rh (D) antigens. A donor's red cells can be given safely, in terms of these antigens, only if the donor carries no antigen that the recipient lacks. This simple rule is easily encoded as a constraint and gives the eight-group compatibility table used in this project (Table 4.1). Final cross-matching is always performed in the laboratory.

## 2.3 Search Algorithms for Routing

Dijkstra (1959) introduced an algorithm that finds the shortest path in a weighted graph by always expanding the node with the lowest cost so far. Hart, Nilsson and Raphael (1968) proposed A*, which adds an admissible heuristic estimate of the remaining distance to the goal. Because the heuristic never overestimates the true cost, A* still returns an optimal path but usually examines far fewer nodes (Russell and Norvig, 2020).

## 2.4 Machine Learning for Donor Availability

Whether a donor responds to an alert depends on factors such as past response behaviour, time since last donation, distance and time of day. Random Forests combine many decision trees trained on random subsets of data and features, which reduces over-fitting and handles mixed numerical and categorical inputs well (Breiman, 2001; Géron, 2022). Such a model can output a probability that a given donor will respond.

## 2.5 Assignment and Language Processing

When several hospitals request blood at the same time, the same donor may be the best choice for more than one request. The Hungarian method (Kuhn, 1955) solves this one-to-one assignment problem optimally. Transformer-based language models (Devlin et al., 2019) can additionally extract the blood group, units and location from a free-text request, reducing data-entry effort.

## 2.6 Research Gap

Existing tools generally address only one part of the problem: a donor directory, a map, or a messaging group. They rarely combine compatibility filtering, eligibility checking, learned availability, route-aware ranking and dispatch tracking in one workflow. The proposed system fills this gap by integrating these steps into a single explainable pipeline.

# 3 PROPOSED SYSTEM

The proposed system is a web and mobile platform that connects hospitals, requesters and donors. A requester enters the patient's blood group, the number of units required, the hospital location and the urgency level. The system then selects, ranks and alerts suitable donors and tracks the outcome.

## 3.1 Architecture Diagram

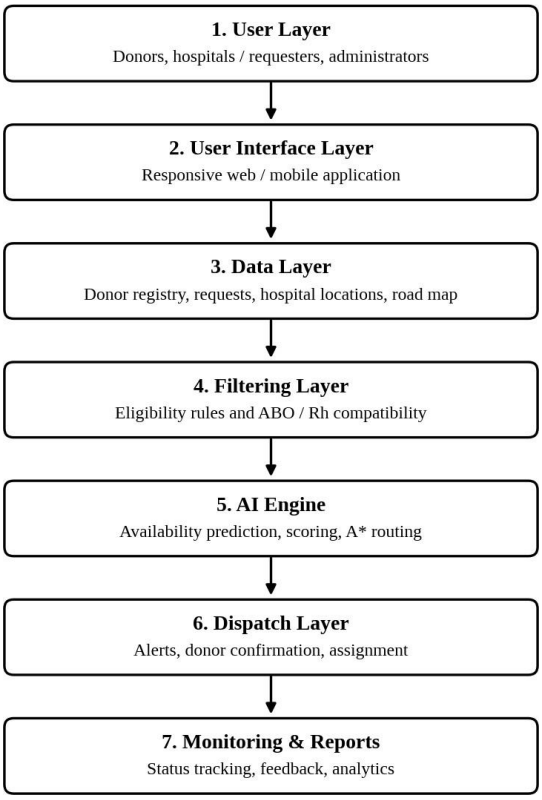



Figure 3.1 Seven-layer architecture of the proposed blood donor matching and dispatch system.

### 3.1.1 Layer Description

The **User Layer** represents donors, hospitals or requesters, and administrators. The **User Interface Layer** is a responsive web and mobile application for registration, request entry and status viewing. The **Data Layer** stores the donor registry, requests, hospital locations and a road map of the service area.

The **Filtering Layer** applies compatibility and eligibility rules. The **AI Engine** predicts availability, computes routes with A* and calculates the weighted score. The **Dispatch Layer** sends alerts, receives accept or decline responses and assigns donors to requests. The **Monitoring and Reports Layer** records outcomes and gives administrators summary reports.

## 3.2 Processing Flow

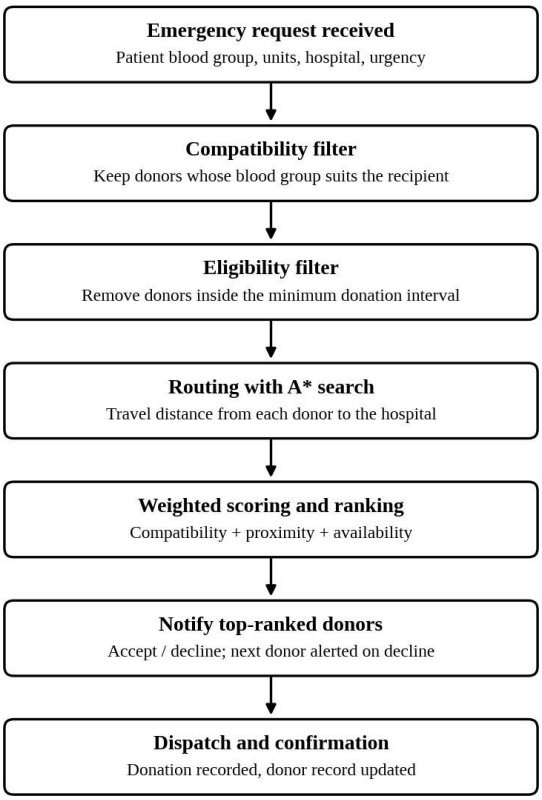



Figure 3.2 Processing flow from emergency request to donor dispatch.

## 3.3 Design Characteristics

- Rule-based safety first: incompatible or ineligible donors are never ranked.
- Route-aware ranking: travel distance is a part of the score, not an afterthought.
- Fallback behaviour: if a donor declines or does not reply, the next donor is alerted automatically.

- Explainable output: every ranked donor is shown with the factors behind the score.
- Human in the loop: hospital staff confirm the final dispatch.

### 3.4 Data and Privacy Considerations

Donor records contain personal and health-related data. The system therefore requires explicit donor consent, shares a donor's contact details with a requester only after the donor accepts an alert, and is intended to follow applicable Indian data-protection requirements, including the Digital Personal Data Protection Act, 2023.

# 4 MODULES DESCRIPTION

## 4.1 Donor Registration and Eligibility Module

This module creates and maintains the donor profile. Each profile stores name, contact number, blood group, location, date of last donation, health declaration and consent. Eligibility is treated as a hard constraint: a donor whose last donation is more recent than the minimum interval is excluded from the candidate list. This project uses an interval of 90 days; the value is a configurable parameter that must follow the guidelines of the local blood bank.

**Output:** a list of eligible donors with their location and history, passed to the compatibility module.

## 4.2 Compatibility Matching Module

Given the recipient's blood group, this module selects the donor groups that can be transfused safely. A donor is compatible when the donor carries no A, B or Rh antigen that the recipient lacks. Table 4.1 lists the resulting rules, which give 27 compatible donor–recipient group pairs out of 64.

Table 4.1 Red-cell compatibility used by the system.

| Recipient group | Compatible donor groups |
|---|---|
| O- | O- |
| O+ | O-, O+ |
| A- | O-, A- |
| A+ | O-, O+, A-, A+ |
| B- | O-, B- |
| B+ | O-, O+, B-, B+ |
| AB- | O-, A-, B-, AB- |
| AB+ | O-, O+, A-, A+, B-, B+, AB-, AB+ |

Compatible donors are graded: an exact group match scores 1.0, another compatible group scores 0.8, and O-negative scores 0.6. The lower grade for O-negative is a design choice that keeps this universal-donor group

available for patients who can receive no other group.

### 4.3 AI Ranking Engine Module

The engine ranks the eligible, compatible donors. For a donor *d* the score is

$$\mathbf{Score(d) = w_1 \cdot C(d) + w_2 \cdot P(d) + w_3 \cdot A(d)}$$

where *C* is the compatibility grade, $P = 1 - (\text{travel distance} \div \text{maximum candidate distance})$ is the proximity score, and *A* is the probability that the donor will respond. The weights satisfy $w_1 + w_2 + w_3 = 1$; this project uses 0.30, 0.40 and 0.30 respectively.

**Availability model.** In the intended deployment, *A* is the output of a Random Forest classifier trained on past alert outcomes, using features such as previous response rate, days since last donation, distance, day of week and time of day. In the prototype, the donor's stored historical response rate is used as a stand-in for this probability, because no real alert history is available.

### 4.4 Routing and Dispatch Module

This module computes the travel distance from each candidate donor to the hospital using A* search over the road network. A* expands nodes in the order of f(n) = g(n) + h(n), where g(n) is the cost from the donor's location to node n and h(n) is an admissible estimate of the remaining distance to the hospital. On a grid road network, the Manhattan distance is such an estimate.

#### Conceptual steps of A*

1. Place the donor's starting node in the open list with f = h.
2. Remove the node with the lowest f value from the open list.
3. If it is the hospital, trace back the parent links and return the path.
4. Otherwise, expand each reachable neighbour and update its cost g if a cheaper route is found.
5. Add the updated neighbours to the open list and repeat from step 2.

For simultaneous requests, the Hungarian method can assign donors to requests so that the total score is maximised without sending one donor to two places. The dispatch logic alerts the top-ranked donors in order, waits for a response and moves to the next donor on a decline or time-out.

### 4.5 Notification and Feedback Module

Alerts are sent to donors through the application and, in a deployed system, through SMS. A donor can accept or decline. After a successful donation, the donor's last-donation date is updated so that the eligibility rule is applied correctly in future. Accept, decline and no-response events are stored, and they form the training data used to improve the availability model over time.

# 5 IMPLEMENTATION AND RESULTS

## 5.1 Experimental Setup

A prototype of the matching, ranking and routing logic was implemented in Python. The experiment was run on synthetic data because no real donor records were available; no real person's data was used.

| Setup element | Description |
|---|---|
| Road network | 24 × 24 grid city (576 cells); 137 cells blocked at random to represent obstacles; each cell is 0.5 km; movement in four directions. |
| Hospital | Fixed at the centre of the grid. |
| Donors | 80 synthetic donors at random free cells, with blood groups drawn from a typical frequency distribution, 20–300 days since last donation, and a response rate between 0.30 and 0.95. |
| Request | Recipient blood group B+. |
| Eligibility | At least 90 days since last donation. |
| Weights | w1 = 0.30, w2 = 0.40, w3 = 0.30. |
| Search | A* with Manhattan heuristic, compared with uniform-cost search (Dijkstra). |

## 5.2 Results

### 5.2.1 Filtering

Of the 80 donors, 54 had a group compatible with a B+ recipient (B+, B−, O+, O−). After removing donors inside the 90-day interval, 37 eligible donors remained, and all of them could reach the hospital on the grid.

### 5.2.2 Ranking and Dispatch Order

Table 5.1 shows the five highest-ranked donors, and Figure 5.1 shows their routes to the hospital. The nearest eligible donor (D33, O+, 0.5 km) was ranked 2nd, not first, because the first-ranked donor has an exact blood-group match and a higher response rate. This shows the difference between ranking by distance alone and ranking by the combined score.

Table 5.1 Top five ranked donors for a B+ request.

| Rank | Donor | Group | Distance (km) | Response rate | Days since donation | Score |
|---|---|---|---|---|---|---|
| 1 | D30 | B+ | 4.0 | 0.88 | 115 | 0.836 |
| 2 | D33 | O+ | 0.5 | 0.57 | 92 | 0.795 |
| 3 | D70 | O+ | 2.0 | 0.67 | 163 | 0.777 |
| 4 | D27 | B+ | 3.5 | 0.62 | 125 | 0.774 |
| 5 | D21 | B+ | 1.5 | 0.34 | 177 | 0.754 |

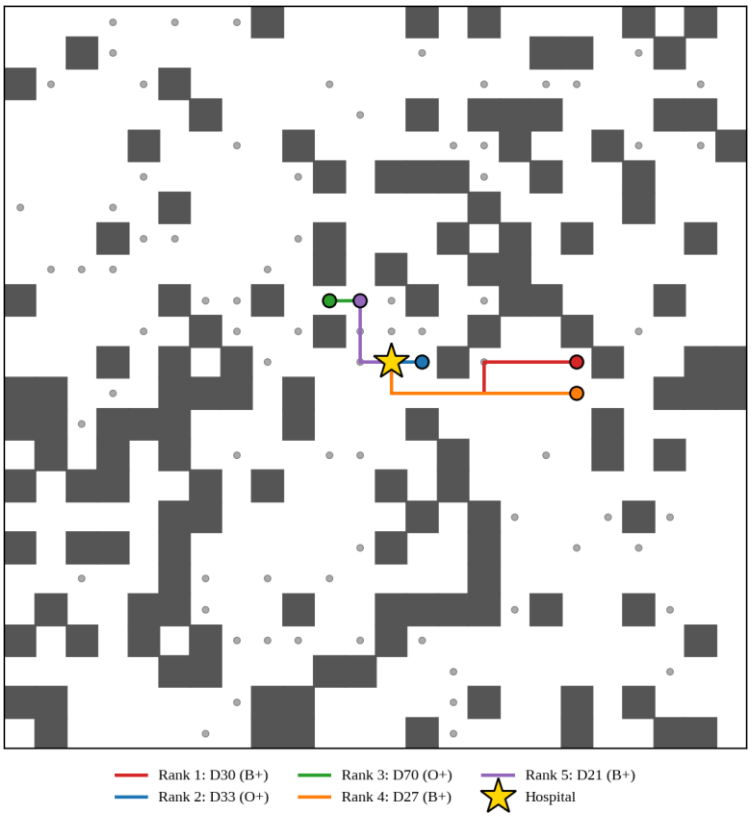



Figure 5.1 Synthetic city grid. Dark cells are blocked; grey dots are donors; coloured lines are A* routes of the five top-ranked donors to the hospital (star).

### 5.2.3 Search Efficiency

For each of the 37 eligible donors, A* and uniform-cost search were run to the hospital. Both returned routes of identical length in every case, as expected for an admissible heuristic. A* expanded 1,232 nodes in total (an average of 33.3 per donor) compared with 4,041 nodes (an average of 109.2) for uniform-cost search, which is about 69.5% fewer node expansions.

These results come from one synthetic grid and one request. They demonstrate that the pipeline works as designed; they are not measurements of real-world response times or clinical outcomes.

## 5.3 Advantages, Limitations and Applications

**Advantages**

- Replaces repeated manual calling with a ranked, rule-checked donor list.
- Protects safety by applying compatibility and eligibility rules before ranking.
- Considers travel distance and likely response together.
- Keeps the scarce O-negative group in reserve where possible.
- Records outcomes so that the availability model can improve.

**Limitations**

- Results are from a synthetic grid; no real donor data, road data or field trial was used.
- The availability model is described but was replaced by historical response rate in the prototype.
- Straight grid distances do not capture real traffic or road conditions.
- The system depends on donors keeping their records and location up to date.
- It does not replace clinical screening, cross-matching or blood-bank procedures.

**Applications**

- Hospital and blood-bank emergency donor search.
- Voluntary blood-donor organisations and campus donor clubs.
- Disaster and mass-casualty response coordination.
- Educational demonstration of search, rule-based reasoning and machine learning.

# 6 CONCLUSION AND FUTURE WORK

## 6.1 Conclusion

This project proposed an AI-based smart blood donor–recipient matching and emergency dispatch system. It combines rule-based compatibility and eligibility filtering, a Random Forest–based availability estimate, A* route search and weighted ranking within a seven-layer architecture. A prototype on a synthetic city grid confirmed that the compatibility rules give the expected 27 compatible group pairs, that ranking by the combined score differs from simple nearest-donor selection, and that A* finds the same shortest routes as uniform-cost search with 69.5% fewer node expansions.

The system is intended as a decision-support tool for faster and better-informed donor selection. It supports, and does not replace, the clinical judgement and procedures of hospitals and blood banks. By joining compatibility rules, machine learning and heuristic search in one workflow, it shows how foundational artificial intelligence techniques can be applied to a time-critical healthcare problem.

## 6.2 Future Work

- Train and validate the Random Forest availability model on real, consented alert-response data.
- Replace the grid with a real road network and live travel-time estimates.
- Add multi-request assignment using the Hungarian method and test it under simultaneous emergencies.
- Integrate SMS or voice alerts and, where permitted, link with national blood-bank portals.
- Add free-text request understanding using a language model.
- Tune the scoring weights from outcome data and run a pilot with a partner hospital.

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